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Batch: 20

Lab report: 10

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Course: CSE LAB

Course Code: 1202

Lab task Topic: Structured Programming Problems Solve {User define function}

Submitted to:

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Problem No. 01

Name of the problem:

Write a program in C to find the square of any number using the function

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//determine square value by user define function
#include <stdio.h>
int main()
{
   int a, result;
   printf("Going to calculate the square value\n");
   printf("Input any number of square:");
   scanf("%d",&a);
   result=sum(a);
   printf("The square of %d is:%d",a,result);
}

int sum(int a)
{
   return (a*a);

}
```

Output:

```
"E:\Programming\my files\C++\Fuction\user_diff_func 1.exe"
Going to calculate the square value
Input any number of square:12
The square of 12 is:144
Process returned 0 (0x0)   execution time : 1.
217 s
Press any key to continue.
```

Problem No. 02

Name of the problem:

Write a program in C to swap two numbers using a function

Introduction

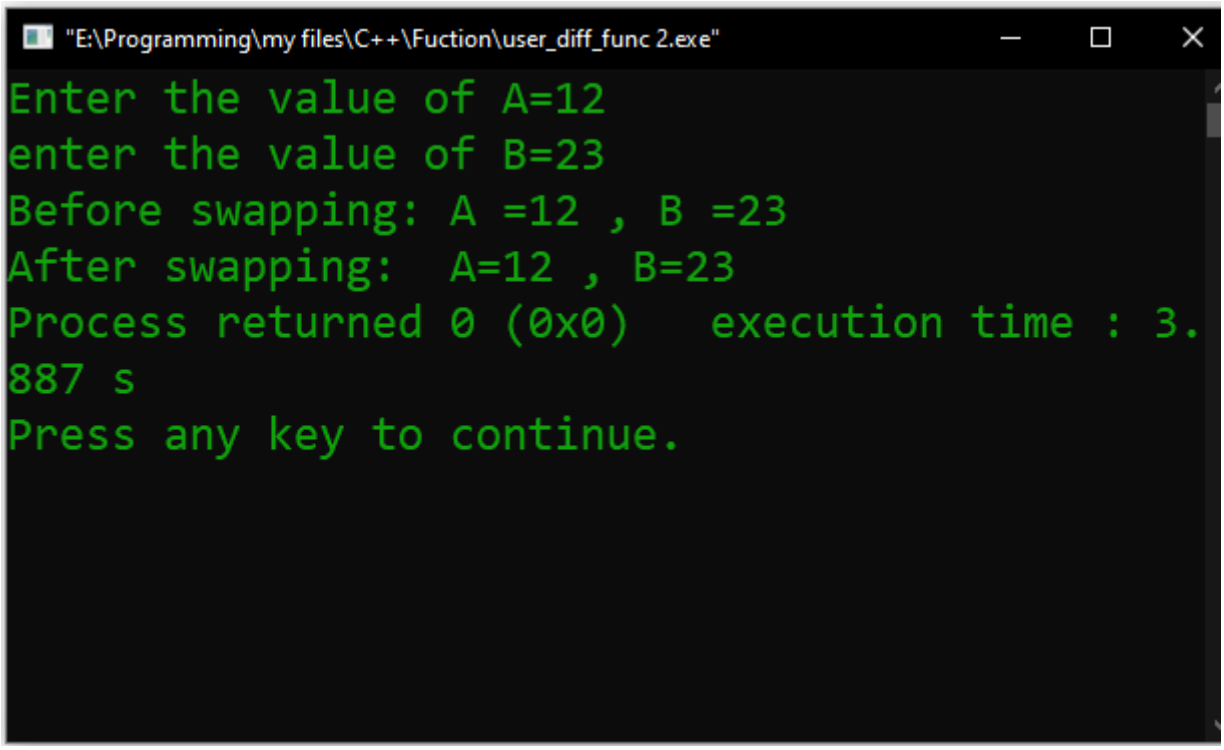
Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//swiping value by user define function
#include <stdio.h>
int main()
{
   int a,b,c,d,e;
   printf("Enter the value of A=");
   scanf("%d",&a);
   printf("enter the value of B=");
   scanf("%d",&b);
   printf("Before swapping: A =%d , B =%d\n",a,b);

   swipe();
   d=swipe(a);
   e=swipe(b);



   printf("After swapping:  A=%d , B=%d", d,e);
}
```

```
int swipe(int a,int b,int c)
{
   c=a;
   a=b;
   b=c;

   return(a,b);
}
```

Output:



Problem No. 03

Name of the problem:

Write a program in C to check a given number is even or odd using the function

Introduction

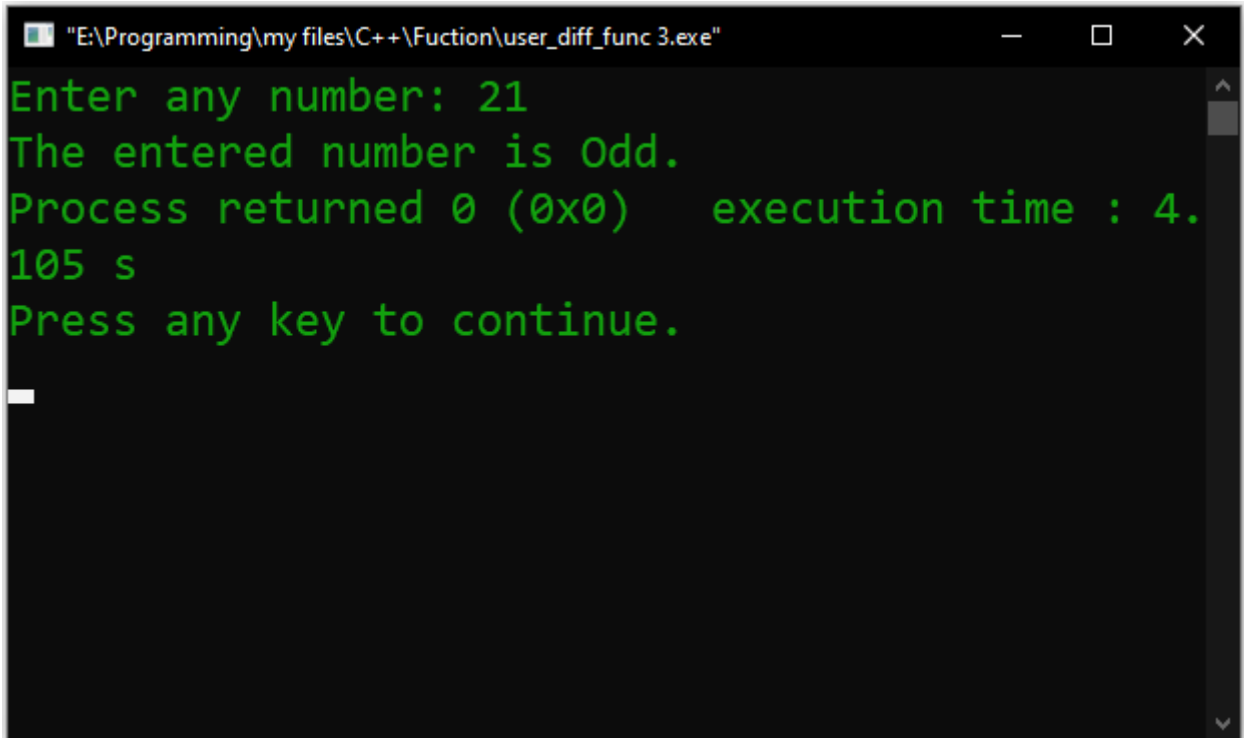
Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//Even Odd (user define function)
#include <stdio.h>
int main()
{
   int a;
   printf("Enter any number: ");
   printf("");
   scanf("%d",&a);

   if(a%2==0)
      even();
   else
      odd();
}
even(int a)
{
   printf("The entered number is Even.");
}
odd(int a)
```

```
{
    printf("The entered number is Odd.");
}
```

Output:



Problem No. 04

Name of the problem:

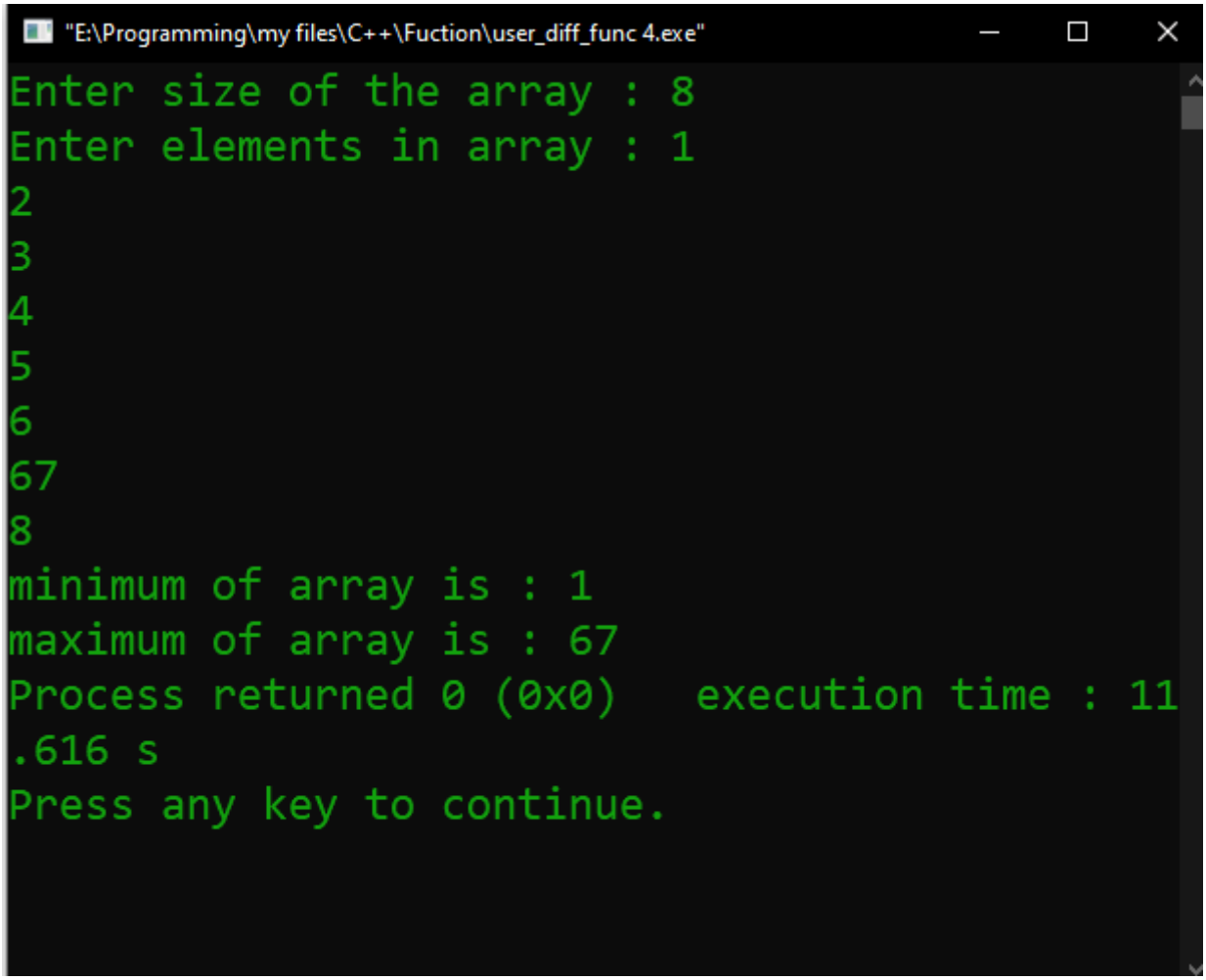
Write a program in C to get the largest element of an array using the function.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//find Maximum value from array numbers user define function
#include <stdio.h>
 int main()
{
   int a[1000],i,n,sum;
   printf("Enter size of the array : ");
   scanf("%d", &n);
   printf("Enter elements in array : ");
   for(i=0; i<n; i++)
   {
      scanf("%d",&a[i]);
   }
   sumofarray(a,n);
}
int sumofarray(int a[],int n)
 {
                int min,max,i;
                min=max=a[0];
   for(i=1; i<n; i++)
```

```
  {
      if(min>a[i])
                min=a[i];
                 if(max<a[i])
                  max=a[i];
  }
 printf("minimum of array is : %d",min);
  printf("\nmaximum of array is : %d",max);
 }
```

Output: